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Construction of complex network of green infrastructure in smart city under spatial differentiation of landscape

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ABSTRACT

In the rapid urban and economic development, environmental problems have become prominent due to urban expansion and land imbalances. However, good historical and cultural landscapes are distributed in urban centers in large cities, and ecologically-friendly forests, lakes, or farmlands are distributed around these cities. It has the spatial differentiation of landscape resources. Landscape resources are used to build a composite green foundation, and the network of facility can improve urban environmental problems. The methods used in the work included landscape connectivity, minimum cumulative resistance model, Gravity model, Kernel Density Analysis, and Circuit theory. Through these methods, the ecological and cultural landscape networks were constructed, forming a composite network of green infrastructure by organic coupling of node composite, construction of composite landscape corridor, and corridor composite. The results showed that the subjective analysis and landscape connectivity analysis were used to quantitatively evaluate the importance of patches, thus selecting the superior ecological nodes. The minimum cumulative resistance analysis and gravity model were used to extract the GI corridors and their importance, respectively, with ecological landscapes constructed. Kernel density analysis and circuit theory could determine the cultural landscape nodes and extract cultural landscape corridors. Node composite, composite landscape corridor construction, and corridor composite are important methods to couple the composite green infrastructures. Compared to traditional ecological network, it can protect regional culture and ecology.

1. Introduction

In China, the high-density urban landscapes and good ecological resources are distributed around cities, while the distinctive and valuable cultural landscape resources are concentrated in urban areas. The distribution of landscape resources has spatial differentiation. These cause the challenges to urban space environment, including population concentration, high intensity of human production and living activities, great ecological demands of urban environment and human, and shortage of urban land resources. Good-quality ecological resources around cities, however, are often ignored. Urban living environment and education of nature and culture can be improved by digging the green resources in high-density urban areas, thus improving green space, service of ecological land, and use of suburban green space.

The concept of green infrastructure proposed in 1990s contains ecological and cultural landscapes [1]. Its essential connotation is to ensure the regional ecological processes and improve the species diversity by constructing green networks, focusing on ecosystem health and landscape resource protection. Methods are provided to improve urban ecological environment [2]. Some scholars used the green infrastructure networks to improve urban ecological service, such as hydrological regulation [3], air conditioning [4], climate regulation [5], and human health [6]. Based on current research, scholars focus on the ecological elements, and the cultural landscape elements are less integrated into network [7,8]. Under the spatial distribution of urban ecological and cultural elements, it is difficult for green networks to enter the high-density urban areas, without improving urban green space. Therefore, under the spatial differentiation of urban landscape, in order to promote urban ecosystems, the work built a network system of green space integrating ecology and culture, namely a complex green infrastructure combining ecological and cultural landscape networks.

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Therefore, under the spatial differentiation of landscape, the construction of composite green infrastructure is divided into the following aspects: The collection and analysis of basic data in the study area include the acquisition and interpretation of remote sensing images, data collection and analysis of urban cultural landscapes, etc. Subjective analysis and connectivity analysis are used to determine the primary and secondary ecological nodes. The minimum cumulative resistance and gravity model are used to extract important ecological corridors to form an ecological landscape network. Kernel density analysis is used to determine the cultural landscape nodes, and extract the cultural landscape corridors through circuit theory, thus building a cultural landscape network. The ecological landscape network is combined with the cultural landscape network through node composite, construction of composite landscape corridor, and corridor composite to obtain a composite network of green infrastructure.

Some scholars use the network of green infrastructure to improve the ecological environment. Liu Xingbo [9] selected 18 protected areas in Shanghai as ecological network nodes according to biodiversity conservation. Wang Haizhen [10] selected 50 green areas with an area greater than 6 ha. as the nodes of ecological network. Kong Fanhua [11] selected 12 green space patches as ecological nodes. Although being convenient and fast, this method is not objective. When the area difference between plaques is not large, it is difficult to judge the quality of plaque. Huang He [12] and Gao Yaling [13] used the area size and connectivity analysis to quantitatively evaluate the importance of ecological plaques, thus determining ecological nodes. Therefore, the work analyzed the connectivity to determine the quality of plaques based on the area and size of plaques. With better ecological plaques selected as nodes, the minimum cumulative resistance model is used to extract ecological corridors and build the networks of ecological landscape. Based on the networks of cultural landscape, Yu Kongjian [14], Li Heping [15] and Wang Sisi [16] used the Delphi method and Kernel density analysis to determine cultural nodes, and the minimum cumulative resistance model and manual selection to extract cultural corridors. A cultural landscape network is established. However, the redundant corridors generated in extracting the cultural landscape corridors by minimum cumulative resistance model cannot be eliminated, which is objective. The circuit theory has been applied to identify the connectivity and construct the corridors, with good applicability in studying green networks [17]. Song Lili [18] used the circuit theory to identify the importance of corridors, which can recognize more important corridors in the network. In addition, Liu Jia [19] used the circuit theory to evaluate the network corridors of green infrastructure in Nanjing. Therefore, the work used the model of minimum cumulative resistance, and the circuit theory to extract important corridors. Then, a network of cultural landscape was established.

The composite green infrastructure network was formed by coupling the networks of ecological landscape and cultural landscape. There is currently no relevant coupling method. The work integrated the cultural and ecological landscape nodes through node composition to build a network of ecological and cultural landscape. The corridors were connected to form a complex network of green infrastructure through integrating corridor.

2. Study-area overview and data sources

2.1. Study-area overview

Fuzhou, capital of Fujian Province, is located on the southeast coast, and its geographical environment is characterized by “leaning against mountain, adjacent to river, and facing the sea”. Jiancheng District has a flat terrain with an altitude of 10–30 m, whose main urban area is the core area of cultural landscapes, especially Wushishan Mountain, Yushan Mountain and Pingshan Mountain. There are land-scale natural scenic areas in suburbs with superior environment, including Lianhuashan Mountain, Gushan Mountain in the north, Qishan Mountan and Wuhushan Mountain in the south. The altitude is about 600–1000 m. It is hot in summer because Fuzhou is located in the mid-south subtropical maritime monsoon climate zone with an average annual rainfall of 900–2100 mm and an average temperature of about 20 °C. Fuzhou is blessed with warm and humid climate, abundant sunshine, lush vegetation and rich species.

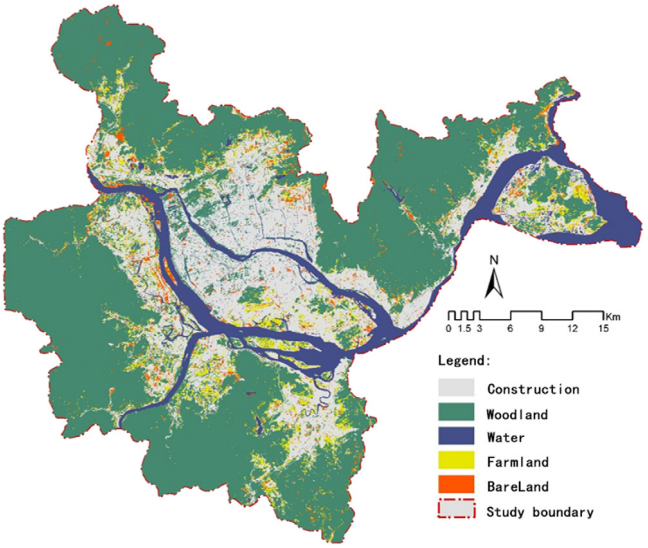


Fig. 1. Land use in Fuzhou.

2.2. Sources and processing of data

The basic data of the work covered the boundaries of Fuzhou, multispectral remote sensing images, protection units of cultural relics, and park squares. The boundary of Fuzhou, as the scope of this study area, was delineated according to the Fuzhou Master Plan (Urban and Rural Planning Bureau) from 2008 to 2020, with a total area of about 1447 km^2. The image data came from the geospatial data cloud website (http://www.gscloud.cn/), of which the Landsat8 multispectral image of Fuzhou in December 2017 was used as the main land-use classification data, and the 30-meter elevation map of GDEMDEM as the topographic landform data. The data of protection units of cultural relics, as the main basis of cultural landscapes, were from the “Fujian Provincial People’s Government Website” and relevant department information (as of December 2018). The data of park squares, roads, and catering facilities were mainly based on Baidu map and relevant government department statistics, while the population data were derived from the Sixth National Census of Population.

ENVI5.1 and GIS software were used to classify the image data into these five types, including construction land (37,004.76 hm^2), bare land (3834 hm^2), woodland (85,816.23 hm^2), water body (13,502.97 hm^2), farmland through supervision and classification (7543.35 hm^2) (See Fig. 1). All the data based on Baidu map was needed to convert into WGS1984 coordinates by projection and transformation, which was used to obtain the data of ecological and cultural landscapes.

3. Research methods

3.1. Method to construct the network of ecological landscape

3.1.1. Selecting ecological nodes

Ecological nodes are selected based on the subjective analysis and quantitative evaluation. Large ecological green areas around cities, such as scenic spots, forest parks and nature reserves, have natural features with less human-induced disturbance, large areas, rich species,

and high levels of protection. In terms of quantity and quality, they can be selected as network nodes, ranking as the first-level ecological nodes. For other green lands, the corresponding quantitative evaluation should be adopted to determine their importance, and some with high importance are chosen as the secondary ecological nodes. The work evaluated the secondary ecological nodes on landscape connectivity, patch area, patch edge length index, and patch importance (dl, Formula 3). Wherein, the patch area and edge length were calculated with GIS software. The high and low energy of landscape connectivity reflected the possibility of species migration in nodes. Moreover, the Conefor2.6 software was used to select four indicators, including possible connectivity (dPC, Formula 1), overall connectivity (dIIC, Formula 2), area and edge width [20], with the secondary ecological nodes determined.

$$PC = \frac{\sum_{i=1}^{n}\sum_{j=1}^{n} a_i a_j p_{ij}^{*}}{A_L^2} \tag{1}$$

$$IIC = \frac{\sum_{i=1}^{n}\sum_{j=1}^{n} \frac{a_i a_j}{1+nl_{ij}}}{A_l^2} \tag{2}$$

$$dI\,(\%) = 100 \cdot \frac{I - I_{remove}}{I} \tag{3}$$

wherein, n is the total number of patches in the landscape; a_i and a_j are the contribution values of patches I and j (referring to area in the work); nl_{ij} is the number of connections between i and j; p^{*}_{ij} the maximum probability that the species spread between i and j; I the landscape connectivity index value; I remove the landscape connectivity value after a patch is removed.

3.1.2. Construction of ecological corridors

(1) Construction of ecological resistance surface

There are resistances in the migration and diffusion of species, whose levels depends on the resistances of different landscapes to species migration [21]. From obstacles affecting species migration and diffusion, the work considered the natural conditions and human-induced disturbance in study area, and analyzed the resistance of different landscapes to species diffusion with expert interviews. For example, in a certain area, green land has a much lower hindrance to species in the area than construction land [22–24].

(2) Construction of potential ecological corridors based on the minimum cumulative resistance model

The minimum cumulative resistance model is used to simulate the optimal path for species migration. The ecological resistance surface of certain area can be used to obtain the sum of ecological resistance of landscape unit that the species passes from source to object point. When the total ecological resistance of a line is the smallest, it is the optimal path to mingrate species [25]. According to this principle, we construct the ecological corridor.

$$MCR = min\sum_{j=n}^{i=m}(D_{ij} * R_j) \tag{4}$$

where MCR is the minimum cumulative resistance value that species need to overcome from source to target point; i and j are different grid units; D_{ij} is the ecological flow, such as traveled distances of material, energy, and information from grid i to the grid j; R_j the weighted resistance value of grid j to ecological flow. The minimum cumulative resistance model reflects the obstruction of ecological flow by different surface characteristics, while the obstruction MCR is positively related to variable distance (D_{ij}) and resistance value (R_j).

(3) Construction of important ecological corridor Based on gravity model

Since potential corridors are iteratively calculated by network nodes, the number of which is large, and not all the plaques need to be connected to each other. Therefore, the evaluation of corridors is required to retain the important corridors. The gravity model is used to quantitatively evaluate the interaction force between source and target. The larger the force is, the more important the corridor is. Based on this principle, the relative importance of the corridor can be evaluated, which contributes to the corridor selection [26]. The gravity model formula is as follows:

$$G_{ab} = k\frac{M_1 M_2}{D_{ab}^{\ 2}} = \frac{\left[\frac{1}{P_a} \times \ln\left(S_a\right)\right]\left[\frac{1}{P_b} \times \ln\left(S_b\right)\right]}{\left(\frac{L_{ab}}{L_{max}}\right)^2} \tag{5}$$

where Gab is the interaction force between the core patch a and b; N_a and N_b are the weight values of two patches; D_{ab} is the normalized value of potential corridor resistance between patches a and b; P_a the resistance value of patch a; S_a the area of patch a; L_{ab} the cumulative resistance value of corridor between patches a and b; L_{max} the maximum value of cumulative resistance of all the corridors in study area.

3.2. Methods to construct the network of cultural landscapes

3.2.1. Selection of cultural landscape nodes based on kernel density analysis

Kernel density analysis, as a common method for the selection of cultural landscape nodes, can be used to calculate the density of evaluation objects in surrounding spatial neighborhood. It is also a widely used non-parametric method in spatial analysis:

$$f\,(x) = \frac{1}{nh}\sum_{i=1}^{n} K\left(\frac{x - x_i}{h}\right) \tag{6}$$

where n is the number of feature objects; f(x) the attribute value of the feature object at x; h the search radius threshold; $x\text{-}x_i$ the threshold from estimated point to sample point x_i; K(·) the kernel function.

Kernel density analysis can be performed by the spatial distance distribution alone, while the attribute value is calculated according to the points of cultural landscape. The results reflect the aggregation of cultural landscape points in spatial distribution [27]. We use the parks, squares, and various levels of cultural relics protection units as basic materials, determining their attribute values according to the elements [15]. For example, the units of national cultural relics protection with high value, far-reaching impact and wide coverage have high attributes, which makes a wide radiation range during spatial analysis. Therefore, the dense cultural landscape points are selected as cultural landscape nodes.

3.2.2. Extraction of cultural landscape corridors based on circuit theory

(1) Construction of cultural landscape resistance surface

The resistance surface of cultural landscape refers to the process and method of constructing ecological resistance surface in constructing the network of ecological landscape; however, it focuses on the resistance of people's culture and eco-tourism recreation [15]. The data of land-use types and elevations in the study area were used to assign values based on the suitability of cultural landscape corridors. Then the grid weighted superposition is used to obtain the resistance surface in the study area.

(2) Extraction of cultural landscape corridors based on circuit theory

Circuit theory, also known as random walk model, is important in probability theory [28]. Doyle et al. believed that the electric charge in the circuit has the characteristic of random walk, which is combined with resistance surface (resistance surface of cultural landscape) to identify the optimal path in landscape [28,29], thus extracting the cultural landscape corridor.

Based on Linkage Mapper, Linkage Mapper, GIS, and Conefor, the work used the Conefor software to calculate the Euclidean distance between cultural landscape nodes. Then the cultural landscape node, the European distance of node, and the resistance surface were processed in the Linkage Pathways Tool under Linkage Mapper (integrated in GIS toolbox).

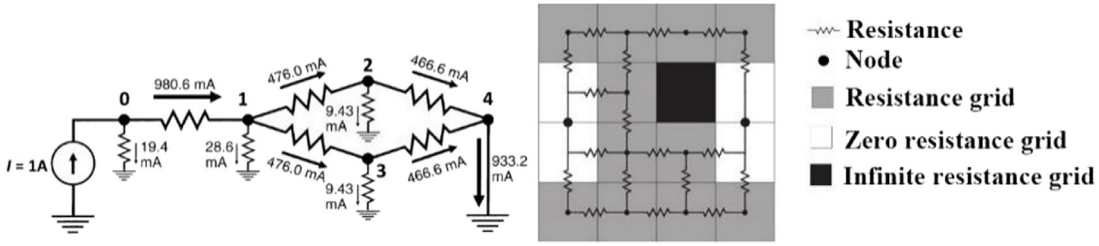


Fig. 2. Landscape raster data structure representation of circuit theory [28].

Table 1
Function of CGIN landscape network.

Composite green infrastructure corridor	Corridor function					
	Species inhabiting	Species migration	Ecological improvement	Recreation	Cultural education	Resource protection
Ecotype	☆☆☆	☆☆☆	☆☆☆	☆	☆	☆☆☆
Culture type	☆	☆☆	☆☆	☆☆☆	☆☆☆	☆☆☆
Compound type	☆☆	☆☆	☆☆	☆☆	☆☆	☆☆☆

Note: More ☆ means more support for this function.

3.3. Construction method of composite green infrastructure network

The difficulty in constructing a complex network of green infrastructure lies in realizing the coupling between ecological and cultural landscape networks. The networks of ecological and cultural landscapes in Fuzhou are constructed through their index evaluation systems, respectively. They cannot satisfy the multiple goals by direct superposition. Moreover, the structure of network is complicated, and the later construction cost is high. Therefore, in order to couple the ecological and cultural landscape networks of study area and achieve the multi-target fusion, the following steps are planned:

(1) Compounding of nodes

The complex green infrastructure network connects the ecological and cultural landscapes, and superimposes ecological and cultural landscape nodes; however, there may be duplicate nodes. If there are both cultural landscape and ecological nodes in the same area, the multiple nodes are integrated into a single node.

(2) Construction of compound landscape corridor

In order to couple the ecological and cultural landscape networks, a "bridge" needs to be set up between two networks. The key point of connection is the nodes of ecological and cultural landscape networks. That is, the nodes after compounding in (1) are connected. Therefore, the Linkage mapper tool is used to connect the ecological nodes with cultural landscape nodes, with the adjacency rule set to 1. The resistance surface uses the results of ecological resistance surface and the cultural landscape resistance surface described above for grid superposition according to the minimum method. This ensures that the corridor routes will choose the units of landscape with low cultural landscape resistance or low ecological resistance.

(3) Integration of corridors

When the composite landscape corridor is in the main urban area, the high-density urban area with tight land resources can be planned as cultural landscape corridor. It not only meets the residents' recreational needs, but also promotes the migration and spread of less sensitive species. According to the characteristics of each landscape network, the functional characteristics of GI composite network are summarized (See Table 1). If there are different types of corridors between two nodes, ecology is the subject to ensure the smoothness of ecological corridors. For subsequent planning, if cultural landscape corridors are required, the planning and construction can be carried out around ecological corridors.

Table 2
Importance evaluation of ecological patches in Fuzhou.

Ranking	Patch No.	dA	dPA	dPC	dIIC	dLCP
1	14	18.007	12.091	39.792	39.098	39.039
2	18	15.266	16.007	26.415	28.102	28.059
3	21	10.460	7.274	16.938	13.192	13.172
4	35	5.558	5.805	3.530	3.725	3.719
5	45	4.828	5.346	2.811	2.811	2.806
6	27	2.857	4.102	1.869	0.984	0.983
7	16	3.385	3.242	1.780	1.452	1.520
8	25	2.830	4.025	2.083	0.966	0.964
9	38	2.614	2.769	1.053	0.824	0.822
10	40	2.074	2.020	0.537	0.519	0.518

4. Results analysis

4.1. Construction of ecological landscape network

4.1.1. Ecological node selection results

(1) Primary ecological nodes. According to the land use classification map, the areas of Lianhua Mountain, Gushan, Qishan, and Wuhu Mountain around Fuzhou are 12,500.6, 16,764.3, 18,278.7, and 18,193.9 hm^2, respectively. As the four largest green space patches in the area and the natural reserve of Fuzhou, the four mountains above own rich species, less man-made interference, and large area. It made them an important ecological source of study area. Selecting four green space patches as primary ecological nodes is conducive to the information interaction and ecosystem health of green space in the area.

(2) Secondary ecological nodes. According to the classification map of land use, after removing the primary ecological nodes, other green space patches with areas more than 10 hm^2 were considered as potential ecological nodes. A total of 57 patches entered the importance evaluation. The area importance (dA), perimeter importance (dPA), possible connectivity importance (dPC), and overall connectivity importance (dIIC) were used for plaque importance evaluation (dI). According to five categories of indicators, the top ranking 10 plaques were chosen as secondary ecological nodes (See Table 2). The rankings are (by number) as follows: 14, 18, 21, 1, 35, 45, 27, 16, 25 and 38. Ten plaques and 4 selected primary ecological nodes constituted the network of green infrastructure in the study area (See Fig. 2).

According to the selection results of primary and secondary ecological nodes, the total area of ecological nodes is 68,000 hm^2, of

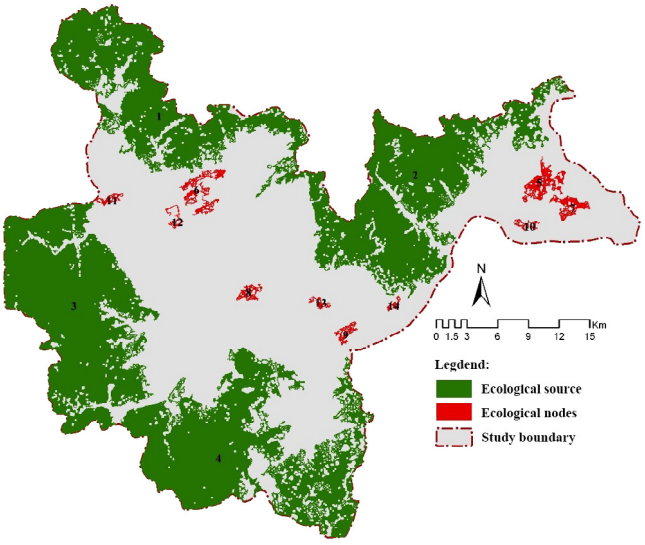


Fig. 3. Map of nodes distribution of ecological landscape network in Fuzhou.

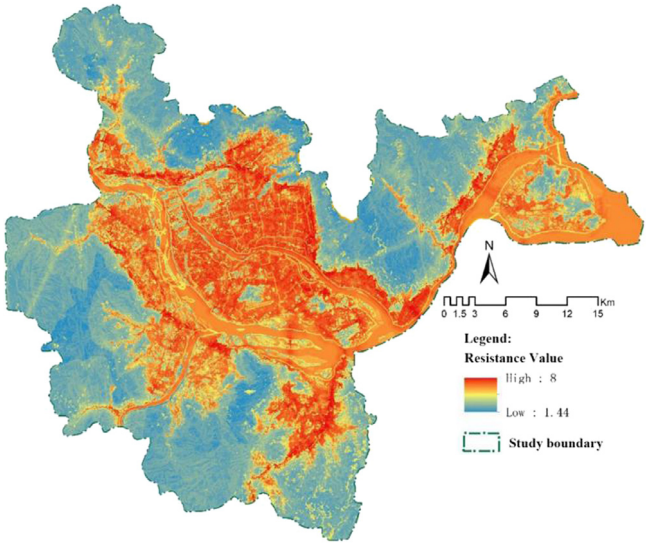


Fig. 4. Fuzhou ecological resistance surface.

which the area of 4 primary nodes is 65,737.5 hm^2 (accounting for 96.6% of the total area of ecological nodes), and the area of 10 secondary ecological nodes is 2262.5 hm^2 (accounting for 3.4% of the total area of ecological nodes). Fig. 3 shows the primary nodes, occupying a large area, are taken as the nodes of large network to study important ecological sources in the study area. Meanwhile, they provide important ecological services such as water conservation and climate regulation. The secondary ecological nodes are distributed in the suburbs and the interior of study area, in which nodes 8 and 12 are distributed in the city, and the remaining nodes are distributed in the suburbs. Various mountain parks belong to the secondary ecological nodes s, which provide not only the ecological services for study area, but also p a place for people to entertain and exercise.

4.1.2. Extraction results of ecological corridors

(1) Construction of ecological resistance surface

Ecological resistance factors are mainly considered from obstacle factors that may affect the distribution and activity of species. For example, the types of land use, slopes, and elevations affect the distribution and migration of species [30]. Higher NDVI values indicate that the green space is healthier and has a positive effect on species diversity. National and provincial forest parks and scenic areas play an important role in restoring and protecting the resources of species. In terms of social disturbance factors, various roads such as high-speed, railway, and national roads have different obstacles to the migration of species. Based on the current status of study area and related literature, a total of 9 resistance factors were selected. Combining the relevant research results and expert consultation, each factor was ranked with 1, 3, 5, 7 and 9. Resistance factors were given corresponding weights according to the expert scoring method (See Table 3). After using the spatial analysis module of GIS to produce the single-factor evaluation results, all the single-factor evaluation raster maps were weighted and superimposed to obtain the ecological resistance value facet map of study area, namely the ecological resistance surface (See Fig. 4).

(2) Results of extracting potential corridors

A corridor generating model based on the minimum accumulated resistance was constructed with the spatial analysis of GIS and the model builder. The model used the primary and secondary ecological nodes as ecological landscape network nodes, and the ecological resistance surface as basic resistance data. Two potential ecological corridors were obtained according to different connections of primary and secondary ecological nodes: 40 primary ecological corridors were extracted by setting 4 primary ecological nodes as the target starting point, 10 secondary ecological nodes as the target end point. Meanwhile, they were connected with the minimum cumulative resistance model. Ten secondary ecological nodes were connected to each other, and a total of 45 secondary ecological corridors were obtained.

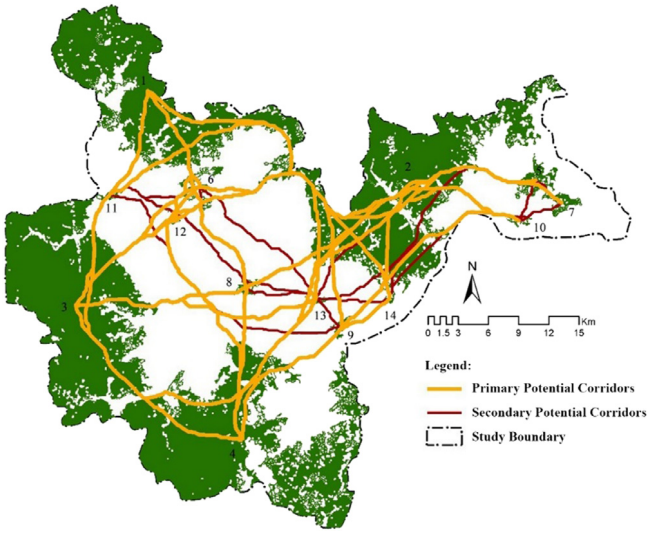


Fig. 5. Potential ecological corridors in Fuzhou.

Fig. 5 shows that the primary potential corridor (yellow line) connects the primary and secondary ecological nodes. The secondary corridor is interconnected between secondary ecological nodes. Moreover, the primary and secondary potential corridors completely cover the study area. The connection between primary and the secondary ecological nodes around the city promotes not only the health of ecosystem in study area, but also the ecological service of internal green space and the species diversity. It is because the internal green space in the city is more closely and frequently connected with the external ecosystem. Wherein, Gulou District, Taijiang District and Jinshan District in the high-density urban area, as well as Shangjie Street and Nanyu on the south bank of Wulong River, have fewer corridors. In the later period, more ecological and green corridors of the area should be constructed to promote the green space quality and ecosystem health.

(3) Extracting important ecological corridors

Due to the large number of potential corridors and complicated routes, the gravity models are used to further screen the potential

Table 3
Evaluation of factors ecological resistance in Fuzhou.

Resistance factor	Weight	Resistance type	Resistance value	Resistance factor	Weight	Resistance type	Resistance value
Elevation	0.08	H ≥ 300 m	1	Primary road buffer	0.06	>800 m	1
		100 < H ≤ 300 m	3			500–800 m	3
		50 < H ≤ 100 m	5			300–500 m	5
		20 < H ≤ 50 m	7			200–300 m	7
		H ≤ 20 m	9			<200 m	9
Slope	0.08	i ≤ 18%	1	Secondary road buffer	0.06	>500 m	1
		18% < i ≤ 30%	3			300–500 m	3
		30% < i ≤ 45%	5			200–300 m	5
		45% < i ≤ 60%	7			100–200 m	7
		i > 60%	9			<100 m	9
Land use type	0.15	Woodland	1	Three-stage road buffer	0.06	>300 m	1
		Farmland	3			200–300 m	3
		Bare land	5			100–200 m	5
		Waterbody	7			50–100 m	7
		Construction land	9			<50 m	9
NDVI	0.1	0.6–1.0	1	Distance from both sides of the water	0.14	≤100 m from main channel	1
		0.4–0.6	3			≤50 m from other kinds of waterbody	3
		0.2–0.4	5			Others	9
		0–0.2	7				
		<0	9				
Natural resource level	0.12	National forest parks	1				
		Provincial forest parks and scenic areas	3				
		Others	5				

Table 4
Interaction matrix between ecological source and ecological nodes based on gravity model (ES refers to ecological source and EN refers to ecological node in Fig. 3)

Patch No.	ES 1	ES 2	ES 3	ES 4
EN 5	20.7	28.2	274.7	37.3
EN 6	165.9	62.8	110.9	643.1
EN 7	17.2	23.0	172.7	29.8
EN 8	134.6	237.2	129.5	92.8
EN 9	54.2	170.9	187.1	46.3
EN 10	16.0	23.2	219.5	29.0
EN 11	485.5	67.1	50.6	384.0
EN 12	221.1	68.3	67.3	278.8
EN 13	61.0	162.8	253.4	63.6
EN 14	30.4	71.5	500.6	57.2

Table 5
Interaction matrix between ecological nodes based on gravity model (EN refers to ecological node in Fig. 3)

Patch No.	EN 5	EN 6	EN 7	EN 8	EN 9	EN 10	EN 11	EN 12	EN 13	EN 14
EN 5	0	34.8	3995.9	37.7	47.6	1572.1	20.3	23.5	53.0	71.6
EN 6		0	27.5	193.4	46.4	27.1	371.5	1968.7	71.0	57.1
EN 7			0	29.5	36.8	1225.3	16.6	18.8	40.0	52.8
EN 8				0	174.2	29.5	84.3	192.6	426.1	90.6
EN 9					0	42.9	28.0	40.1	1338.3	515.5
EN 10						0	15.8	18.3	43.0	69.0
EN 11							0	446.9	38.4	27.4
EN 12								0	62.3	35.4
EN 13									0	341.8
EN 14										0

corridors. The gravity model can evaluate the strength of interaction force between nodes. The greater force means that there is the biological communication between two nodes, which is used to evaluate the importance of corridor. The connection between primary and secondary ecological nodes is used to evaluate the importance of primary and secondary potential corridors (See Tables 4 and 5).

Table 4 shows the results of evaluating primary potential corridors. A total of 10 corridors are selected as the primary corridors with an interaction force greater than 200. Among them, each of the primary and secondary ecological nodes has 1 or more corridors connected, and node 3 has 4 corridors (the most). Nodes 4 and 6 own the highest interaction forces in the primary corridors, followed by nodes 3 and 14, and 1 and 11. This indicates the high possibility for the species information, matter, and energy interaction between these paired plaques. Therefore, the corridors should be constructed to ensure their connectivity. Nodes 1 and 10, 1 and 7, 2 and 7, and 2 and 10 have smaller interaction forces, indicating that the possibility of species exchange between them is low. Considering the construction of corridors between these nodes is costly and inefficient, the corridors in between are not required to be retained.

Table 5 shows the results of evaluating the interaction forces of secondary potential corridor. The secondary corridors with a force greater than 100 with a total of 13 are selected. Node pair 5 and 7 with closer distance and relatively large area has a large interaction force, followed by node pairs 6 and 12, 5 and 10, and 9 and 12.

A network of ecological landscape in the study area was constructed based on 14 primary and secondary ecological nodes. Besides, 22 primary and secondary ecological corridors were extracted by the minimum cumulative resistance model and gravity model. Fig. 6 shows that the network nodes of ecological landscape and the corridors are mainly distributed in the urban periphery area. There are secondary corridors along the Minjiang River and Wulong River, but few in the high-density urban areas such as Gulou District, Taijiang District, Jin'an District and Cangshan District. Hence, more green spaces and ecological corridors in these areas should be constructed in the later stage, which is conducive to their ecological service.

4.2. Cultural landscape network construction results

4.2.1. Results of selecting cultural landscape node

The collected cultural landscapes were assigned 1–4 points based on their ecological value, cultural value, and service scope [27]. Wherein, 4, 3, 2 and 1 were assigned to the protection units of national cultural

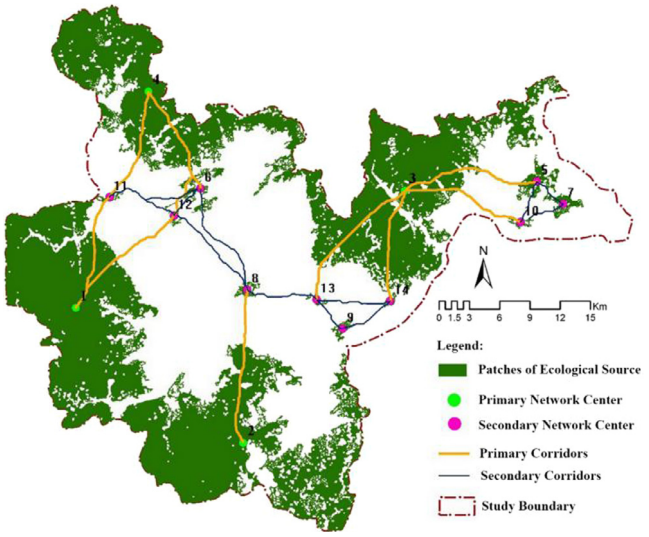


Fig. 6. Network of ecological landscape in Fuzhou.

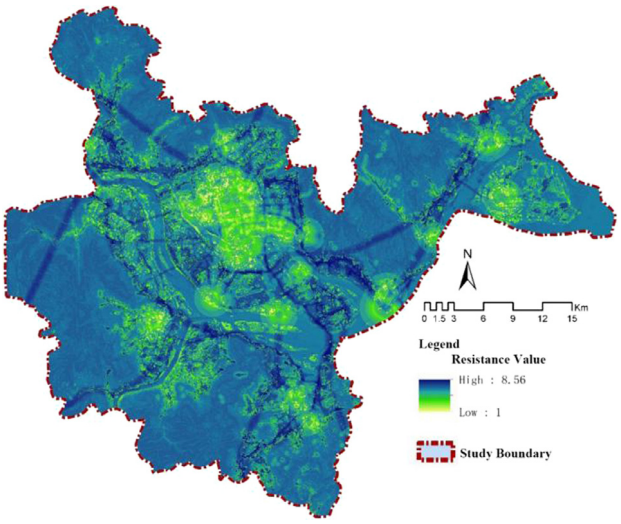


Fig. 8. Resistance surface of cultural landscape in Fuzhou.

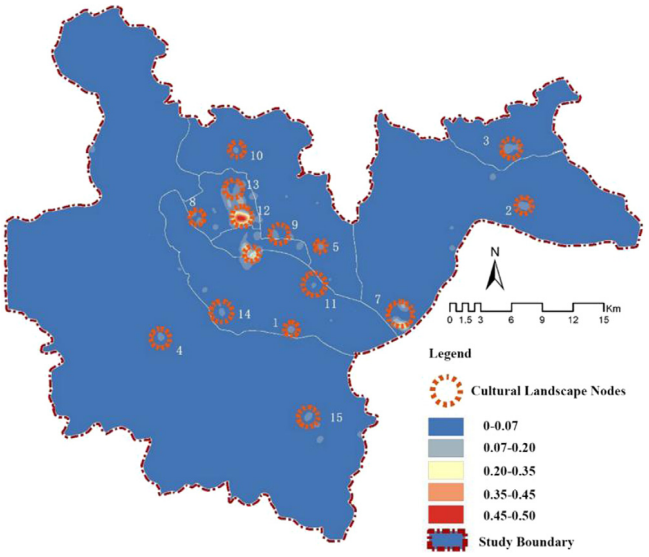


Fig. 7. Spatial distribution map cultural landscape nodes in Fuzhou.

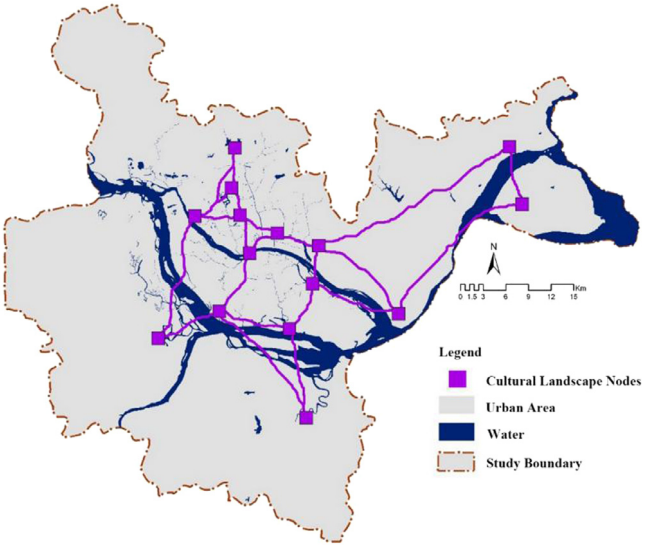


Fig. 9. Cultural landscape network in Fuzhou.

relics and the parks, provincial cultural relics protection units, municipal cultural relics protection units, and squares, respectively. Using the module of GIS kernel density analysis, the distribution of cultural landscape aggregation in the study area was obtained, and 15 cultural landscape aggregation areas were selected as cultural landscape nodes. Fig. 7 shows that the cultural landscape areas are concentrated in Gulou and Taijiang districts, where Sanfang Qixiang Historic District in Gulou District has the highest concentration, followed by Shangxianghang Historic District in Taijiang District. Cultural landscape nodes are mainly distributed in high-density urban areas, and less distributed around the city.

4.2.2. Results of extracting cultural landscape corridors

(1) Constructing the resistance surface of cultural landscape

Cultural landscape corridors, the connecting lines connecting different nodes of cultural landscape, are convenient places for urban residents to travel, exercise, and relax between cultural landscape nodes. Meanwhile, it has ecological benefit, providing the channels to migrate and diffuse species as urban greenways. The resistance factors of cultural landscape are determined based on socio-economy, aesthetics, landscape preference, recreational needs and ecology. According to the current situation and related literature of the study area, a total of 11 resistance factors were selected, and the resistance values and weights were then obtained in terms of the assignment of ecological resistance factor and the weight scoring methods (Table 6). Using the GIS spatial analysis function, each resistance factor is weighted and superimposed to obtain the resistance surface of cultural landscape in the study area (See Fig. 8).

(2) Extracting the corridors of cultural landscape

Linkage pathway tool of Linkage Mapper was used for extraction. The minimum connecting corridors between nodes were 1 and 2, respectively, and there are 22 cultural landscape corridors in all (See Fig. 9). When the minimum number of connections is set as 1, the constructed corridors are most important in the cultural landscape network, classified as the corridors of primary cultural landscape in the study area. When the minimum number of connections is set as 2, 8 corridors (excluding the primary corridors) are constructed, defined as the corridors of secondary cultural landscape in the study area (See Fig. 9).

Table 6
Evaluation of factors ecological resistance in Fuzhou.

Resistance factor	Weight	Resistance type	Resistance value	Resistance factor	Weight	Resistance type	Resistance value
Elevation	0.049	H ≤ 20 m m20 < H ≤ 50 m 50 < H ≤ 100 m 100 < H ≤ 300 H ≥ 300	1 3 5 7 9	Distance of two sides of waterbody	0.082	D ≤ 50 m 50 < D ≤ 100 m 100 < D ≤ 200 m 200 < D ≤ 300 m D > 300 m	1 3 5 7 9
Slope	0.049	i ≤ 5% 5% < i ≤ 10% 10% < i ≤ 25% 25% < i ≤ 50% i > 50%	1 3 5 7 9	Cultural landscape value	0.082	≥0.45 0.35–0.45 0.2–0.35 0.07–0.2 <0.07	1 3 5 7 9
Land use type	0.15	Woodland, farmland Bare land Waterbody – Construction land	1 3 5 7 9	Cultural landscape gathering area buffer	0.167	300 m 300–600 m 600–900 m 900–1500 m 300 m	1 3 5 7 9
Railway, freeway buffer	0.067	>500 m 400–500 m 300–400 m 200–300 m <200 m	1 3 5 7 9	Population distribution	0.035	>0.3 0.02–0.03 0.01–0.02 0.04–0.01 <0.04	1 3 9
National road, provincial road buffer	0.067	>200 m 150–200 m 100–150 m 50–100 m <50 m	1 3 5 7 9	Catering service	0.035	>1.7 1.2–1.7 0.7–1.2 0.2–0.7 <0.2	
Urban road buffer	0.067	>100 m 100–80 m 50–80 m 20–50 m <20 m	1 3 5 7 9				

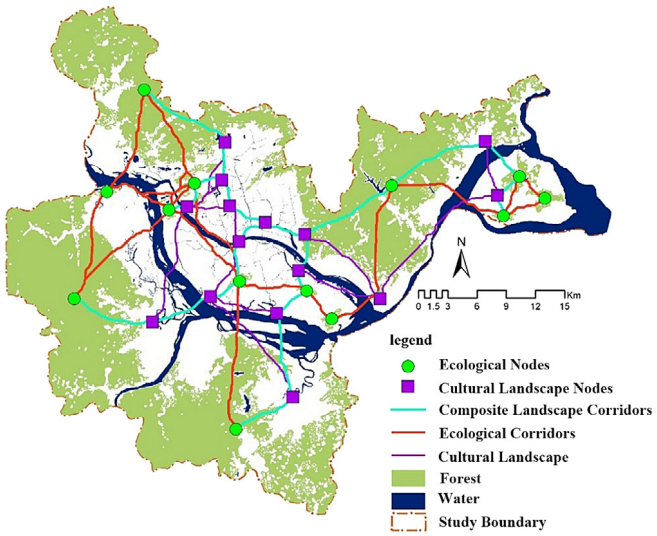


Fig. 10.

4.3. Results of constructing the complex network of green infrastructure

(1) The Integration of nodes. Comparing the results of constructing the networks of cultural landscape and ecological landscape, 14 ecological nodes and 15 cultural landscape nodes were founded in all. From their spatial distribution, ecological node 29 and cultural landscape node 7 were overlapped, and a total of 28 landscape nodes were left after the two merged.

(2) Constructing the resistance surface of composite landscape. The ecological resistance surface and the resistance surface of cultural landscape were used as basic data, while the GIS minimum method was used for spatial superposition to obtain the resistance surface of composite landscape.

(3) Constructing composite corridors. Linkage Mapper was used to connect ecological nodes and cultural landscape nodes (the adjacency rule is 1) to build the necessary "bridge" between the networks of cultural and ecological landscape, which was the composite corridor.

(4) Integrating the corridors. Three types of structure, ecological corridors, cultural landscape corridors and composite corridors, constituted a composite green infrastructure network, but there were duplicate or redundant corridors, which should be integrated. According to the functional positioning of the corridor types in Table 2, the work decided to first preserve ecological corridors as far as possible. After saving the composite landscape corridors, the cultural landscape corridors were analyzed to integrate three corridors.

After steps 1, 2, 3, and 4, a composite network of green infrastructure in the study area was constructed (See Fig. 10), which composed of ecological landscape network, cultural landscape network and composite landscape corridor. Fig. 10 shows that the ecological and cultural landscape networks are coupled to form a composite network of green infrastructure. There are 14 ecological nodes and 15 cultural landscape nodes. After eliminating duplicate nodes, there are 28 composite green infrastructure network nodes left. Three types of corridors are formed, with a total of 54. The total length is 432.6 km—19 ecological corridors with a total of 196.5 km, 12 cultural landscape corridors with a total of 110.8 km, and 23 composite landscape corridors with a total of 125.3 km. Many natural corridors are among the ecological corridors. For example, the three corridors in the Lianhua Mountain Scenic Area (the northwest of study area) are mostly natural forest lands. In addition, the Qishan area, Wuhushan area, and Gushan area are large natural green spaces, formed natural corridors within them are all.

Table 7
Comparison of distribution of the composite network of green infrastructure and the network of traditional green space.

Network structure	Complex green infrastructure in study area	Ecological network in Shang Hai [9]	Ecological network in pingtan island [12]	Network of green infrastructure in Fuzhou [13]
Number of nodes in the city	16	3	0	4
Number of suburban nodes	12	15	10	9
Number of corridors in the city	28	0	2	2
Number of connecting corridors inside and outside the city	14	3	0	3
Number of suburban corridors	18	7	16	5

4.4. Comparing the composite networks of green infrastructure with traditional greenfield networks

Compared with the traditional network of green infrastructure (See Table 7), the composite network of green infrastructure constructed in the work has a spatial distribution covering entire research area, and integrates urban landscape resources. Table 7 shows that the network nodes of composite green infrastructure in this study are more reasonably distributed in the city and suburbs, with 16 and 12 nodes, respectively. There are 28 corridors in the city—14 corridors connecting the city and outside, and 18 corridors in the suburb Road, covering the urban area. The Pingtan ecological network [28] has no nodes in the city, and only two corridors pass through the city. The remaining nodes and corridors are distributed in the suburbs, where the ecological network is mainly distributed. While the ecological network in Shanghai and the green infrastructure network in Fuzhou have a small number of nodes or corridors in city branch, the green space network of large branch is distributed around the city, which has few effect on improving the green space within the city.

Integrating a composite network of green infrastructure of high-density urban landscape nodes into the green infrastructure network not only protects the ecological diversity of green spaces around the city and the health of ecosystem, but also improves the health and service capabilities of green space within the city. On the one hand, the composite network of green infrastructure covers the city scope, allowing the network of green infrastructure to enter high-density urban areas. On the other hand, main functions of ecological and cultural landscape network are classified, which is beneficial to the interaction of "landscape flow" in green infrastructure network.

5. Conclusions

The work constructed a composite green infrastructure network coupling the ecological and cultural landscapes of Fuzhou based on the current status of spatial distribution of landscapes common in large and medium-sized cities in China. Meanwhile, using theories of landscape ecology and heritage corridors as the foundation, the construction of the network involved connectivity analysis, minimum cumulative resistance model, nuclear density analysis, and circuit theory. The conclusion is as follows:

Four primary ecological nodes and 10 secondary nodes selected by subjective evaluation and quantitative evaluation methods were all distributed around the city or the edge of the main urban area. The quality of green space in high-density urban areas was poor, difficult to enter the ecological landscape network. In the later ecological construction process, attention should be paid to improving the internal green space quality. Measures should be adopted, including expanding the green space of city center park, improving its connectivity and reducing human interference.

Twenty-two ecological corridors were extracted by using the minimum cumulative resistance model and the gravity model. Most of corridors were located on the edge of city, indicating that the ecological suitability of high-density urban areas was poor, and ecological construction should be strengthened. The constructed ecological landscape network is basically distributed around the city. The main reason is that the theoretical background of ecological landscape network construction is landscape ecology, with the poor quality of green space in high-density urban areas. Compared with the urban ecological conditions, no ecological nodes or corridors are found suitable. For the urban ecological environment by using the networks of green infrastructure, we should improve the green spaces around cities, especially the internal green spaces distributed around people's work and life.

Kernel density analysis was used to select 15 hotspots as cultural landscape points. Most of them are located in the city in Gulou District, Taijiang District, and Jin'an District. These three areas are in the ancient city of study area, which accords with the humanity of study area. The circuit theory was deployed to extract primary and secondary cultural landscape corridors, which were mainly distributed in high-density urban areas. Primary cultural landscape corridors were taken as the most important channels between nodes in their construction and protection. Secondary cultural landscape corridors were complement to the primary corridor. The cultural landscape network was constructed based on the heritage corridor and green infrastructure theory. Under the landscape protection, it should not only provide a place for people to enjoy leisure sports and culture, but also enhance the ecological construction of nodes and corridors. Appropriate sites ought to be offered for species to inhabit, migrate and diffuse.

The ecological and cultural landscape networks were superimposed, and the composite network of green infrastructure was coupled through node integration, construction of composite landscape corridors, and corridor integration. There were 28 ecological and cultural landscape nodes, and 54 ecological, cultural, and composite landscape corridors. Their spatial distribution covered the entire study area, and integrated the resources of urban landscape.

The integration of high-density urban landscape nodes into the green infrastructure network will not only be beneficial to the protection of the ecological diversity of green spaces around the city and the health of the ecosystem, but will also promote the health and service capabilities of green space within the city. On the one hand, the composite green infrastructure network comprehensively covers the city scope, allowing the green infrastructure network to enter high-density urban areas. On the other hand, its clarification of main function of the ecological and cultural landscape network is beneficial to the interaction of "landscape flow" in the green infrastructure network.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

He Huang: Conceptualization, Methodology, Investigation, Writing - original draft. **Minghuan Zhang:** Data curation, Writing - original draft. **Kunyong Yu:** Resources, Software, Validation. **Yaling Gao:** Formal analysis. **Jian Liu:** Supervision, Project administration.

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